Similar engagement of the production system in L1 and L2 prediction

Introduction: Language prediction has been proposed to rely on the production system, with comprehenders recruiting production mechanisms to anticipate upcoming words [1]. In fact, evidence shows that blocking the production system during reading reduces prediction effects, indicating that production is involved in prediction [2]. This raises an important question for second language (L2) comprehension, where prediction is typically weaker than in a first language (L1) [3]. Because L2 production is generally slower and less fluent, prediction-by-production may be less efficient, or even not viable, in an L2 [1]. If L2 prediction does not rely on production, blocking the production system should have little or no impact. The present study provides the first direct test of whether L2 prediction relies on production, as L1 prediction does.

Methods: We tested 32 native speakers of Spanish who had learned French later in life. Each participant completed a replication of Martin et al. [2] in both Spanish (L1) and French (L2). Participants read sentences word by word while EEG was recorded. Each sentence contained a highly constraining context that ended in either an expected or an unexpected noun-phrase. We manipulated the availability of the production system through two within-participant conditions: a condition in which participants repeatedly produced a syllable while reading, blocking their language production system (syllable production condition); and a control condition in which participants only tapped their tongue while reading, not blocking their language production system (tongue tapping condition). Event-Related Potential (ERP) mean amplitudes were calculated in the N400 time window (250-450 ms), time-locked to the onset of articles and nouns of the critical noun-phrases, to assess the typical N400 prediction effect, i.e., more negative amplitudes for unexpected compared to expected continuations.

Results: In both languages, blocking the production system reduced N400 prediction effects at the noun level. In Spanish (L1), there was a significant N400 effect ($p < .001$), and a significant interaction with condition ($p = .001$) (Figure 1; time windows of interest highlighted in yellow). Namely, the N400 effect was smaller during syllable production than during tongue tapping. We also observed a significant late anterior positivity (LAP; 600-700 ms time window): unexpected nouns elicited more positive amplitudes than expected nouns ($p < .001$), and this effect also interacted with condition ($p = .005$), being smaller during syllable production than during tongue tapping. Importantly, French (L2) showed the same N400 pattern: a significant N400 effect ($p < .001$) and a significant interaction with condition ($p < .001$), showing a smaller N400 effect during syllable production than during tongue tapping (Figure 2). Unlike in L1, no significant LAP effect was observed in the L2. No significant N400 prediction effects emerged on the article in either language (ps $> .05$) (Figures 3 and 4).

Conclusion: We observed reduced N400 prediction effects when the participants' production system was blocked (syllable production) compared to a control task (tongue tapping). Crucially, this reduction was similar in both L1 (Spanish) and L2 (French), providing the first direct evidence that L1 and L2 prediction rely on production mechanisms to the same extent. The manipulation also reduced LAP effects in L1, whereas no LAP was observed in L2, in line with previous research [4]. In contrast with the original study [2], no prediction effects were found on article onset for any language, making it difficult to determine whether the observed effects at the noun reflect predictive processing, integration, or both. Nevertheless, and more importantly, blocking the production system modulated ERP responses in a similar manner in both L1 and L2. Rather than supporting the view that L2 prediction is limited by production efficiency, these results point to shared predictive mechanisms across L1 and L2. This provides new insight into the mechanisms underlying prediction and L1-L2 differences and similarities in this function.

Figure 1. Mean amplitude in central electrodes (Cz, C3, FC5, CP5, T7, C4, FC6, CP6, T8) elicited by nouns in Spanish (L1)

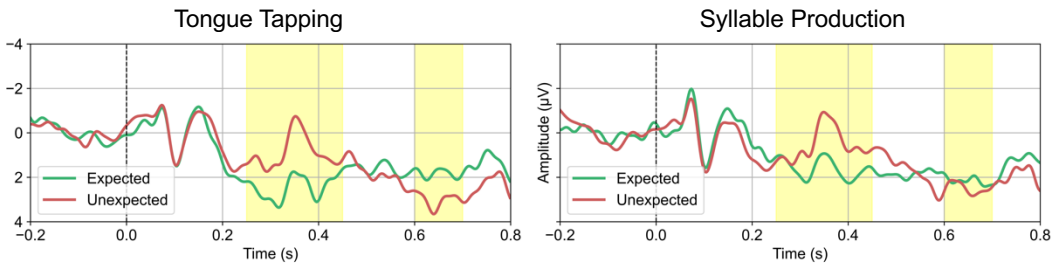


Figure 2. Mean amplitude in central electrodes (Cz, C3, FC5, CP5, T7, C4, FC6, CP6, T8) elicited by nouns in French (L2)

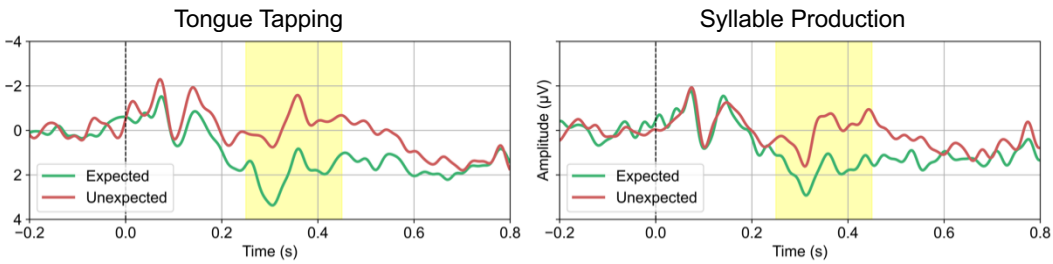


Figure 3. Mean amplitude in central electrodes (Cz, C3, FC5, CP5, T7, C4, FC6, CP6, T8) elicited by articles in Spanish (L1)

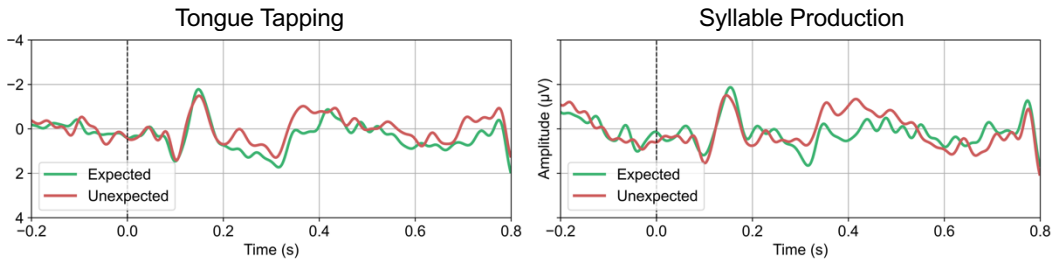
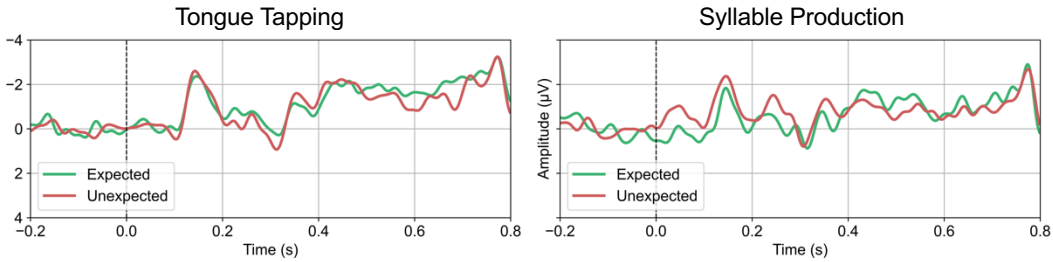


Figure 4. Mean amplitude in central electrodes (Cz, C3, FC5, CP5, T7, C4, FC6, CP6, T8) elicited by articles in French (L2)



[1] Pickering & Gambi (2018). *Psychol Bull*. [2] Martin et al. (2018). *Sci Rep.* [3] Schlenter (2023) *Biling-Lang Cogn*. [4] Martin et al. (2013) *J Mem Lang*.